

Implementation / Feasibility Plan

Feasibility conclusion

Yes. Evidence Locker can realistically be implemented as a pilot and matured into a production service. The repository already contains a working web architecture, backend routes, database models, authentication, evidence handling, PDF generation, and an AI chatbot integration.

Feasibility rating

Technical feasibility: High for pilot | Operational feasibility: Moderate to high with training and ownership | Production readiness: Requires hardening, monitoring, privacy governance, and controlled rollout.

Implementation principle

- Start with a controlled pilot for complaint preparation and evidence organization.
- Validate workflow quality and user trust before expanding institutional integrations.
- Treat AI guidance as an assistive feature with human verification and clear failure handling.
- Harden security, privacy, reliability, and operations before production scale.

1. Current Implementation Baseline

The repository demonstrates a coherent foundation rather than only a conceptual design. This reduces implementation risk because the principal application surfaces and data flows are already represented in code.

Frontend

React, Vite, routing, responsive UI, complaint and evidence screens, authenticated chatbot surface.

Backend

Node.js and Express APIs for auth, complaints, evidence, audit, jurisdiction, and chat.

Data and evidence

MongoDB/Mongoose models, GridFS storage utility, Multer validation, metadata extraction, and SHA-256 hashing.

Documents and controls

pdf-lib complaint package generation, OTP/JWT sessions, RBAC, encrypted contact fields, audit logs, and retention support.

Why this is feasible

- The solution uses established, widely supported technologies and standard web deployment patterns.
- The principal workflows have clear boundaries between UI, API routes, models, evidence utilities, and document generation.
- The AI chatbot is isolated behind an authenticated route and can fail explicitly when Gemini is unavailable or unconfigured.
- The product can begin without complex external integrations; official integrations can be added only after authorization and privacy review.

2. Required Implementation Environment

Technical requirements

- Managed Node.js runtime for the Express backend and a static hosting or CDN layer for the frontend.
- Managed MongoDB with encrypted connections, access controls, backups, and tested restore procedures.
- Durable evidence storage through GridFS or an approved object-storage equivalent with lifecycle controls.
- HTTPS, secure secret storage, domain configuration, logging, monitoring, alerting, and rate-limit protection.
- SMTP or transactional email provider for OTP delivery and a secured Gemini API configuration for chatbot features.

People and operating model

- Product owner responsible for requirements, scope, and acceptance criteria.
- Backend/frontend maintainers responsible for releases, defects, and security updates.
- Support or NGO operators trained on complaint intake, evidence context, privacy, and escalation.
- Designated administrators responsible for access reviews, retention, incident response, and audit review.
- Institutional or legal reviewers for policies, official handoffs, Section 65B use, and jurisdiction-specific processes.

Feasibility dependency

Technology alone is not sufficient. Clear ownership, training, privacy rules, and an escalation process are required for reliable operation.

3. Phased Delivery Plan

A staged rollout limits risk and creates evidence that the product is useful before broader deployment.

Phase 1: Pilot

Deploy for a small group of victims and support workers. Validate intake, evidence upload, chatbot guidance, PDF output, and basic audit workflows.

Phase 2: Hardening

Add production secrets management, HTTPS, backups, monitoring, rate limits, access reviews, retention procedures, and security testing.

Phase 3: Institutional trial

Introduce selected police, NGO, or institutional reviewers with defined roles, training, handoff rules, and support ownership.

Phase 4: Scale

Optimize storage and performance, expand language/accessibility support, and add approved integrations after governance review.

Pilot acceptance criteria

- Users can register, authenticate, create complaints, and recover from common validation errors.
- Evidence can be uploaded, linked, hashed, described, reviewed, and included in a generated package.
- Authorized users can see appropriate records while unauthorized access is rejected.
- The chatbot provides useful guidance, handles unavailable AI service conditions, and does not expose secrets or unsupported claims.
- Operators can explain retention, escalation, privacy, and official-process boundaries to users.

4. Risks and Mitigations

Risk	Mitigation
Sensitive data exposure	HTTPS, secret management, encryption, RBAC, access reviews, limited retention, and incident response.
AI hallucination or overreliance	Bounded system instruction, metadata-only context, human verification, explicit limitations, timeout and error handling.
Evidence loss or corruption	Durable storage, SHA-256 hashes, backups, restore tests, upload validation, and custody records.
Low user adoption	Simple language, guided forms, multilingual support, support-worker training, and usability feedback.
Unclear institutional ownership	Named administrators, documented escalation paths, role definitions, and pilot agreements.
Scale and availability	Managed services, monitoring, rate limits, performance testing, and staged rollout.

Key implementation caution

Do not treat the current repository as automatically production-ready. The implementation is a strong pilot foundation, but deployment must include security review, privacy controls, operational monitoring, backup validation, and user acceptance testing.

5. Feasibility Judgment and Next Steps

The solution can realistically be implemented because the main product workflows and technology foundation already exist. The remaining work is primarily production hardening, operational definition, testing, training, and controlled adoption rather than fundamental research.

Overall feasibility: REALISTIC WITH CONTROLLED DELIVERY

Pilot feasibility is high. Production feasibility is achievable after security, privacy, reliability, governance, and training requirements are completed.

Recommended next steps

- Freeze a pilot scope and define target users, supported complaint categories, and success measures.
- Run end-to-end tests for authentication, authorization, evidence upload, hashing, custody, PDF generation, and chatbot failure paths.
- Complete a privacy and security review covering secrets, sensitive fields, evidence storage, retention, logs, and AI data flow.
- Deploy with backups, monitoring, alerting, rate limits, and a tested incident-response process.
- Train support workers and administrators, then collect pilot feedback before institutional expansion.
- Add external integrations only after formal authorization, data-sharing review, and operational ownership are established.

Final conclusion

Evidence Locker is technically and operationally feasible as a staged implementation. Its success depends less on inventing new core technology and more on disciplined deployment, responsible AI use, secure handling of sensitive evidence, and sustained institutional ownership.